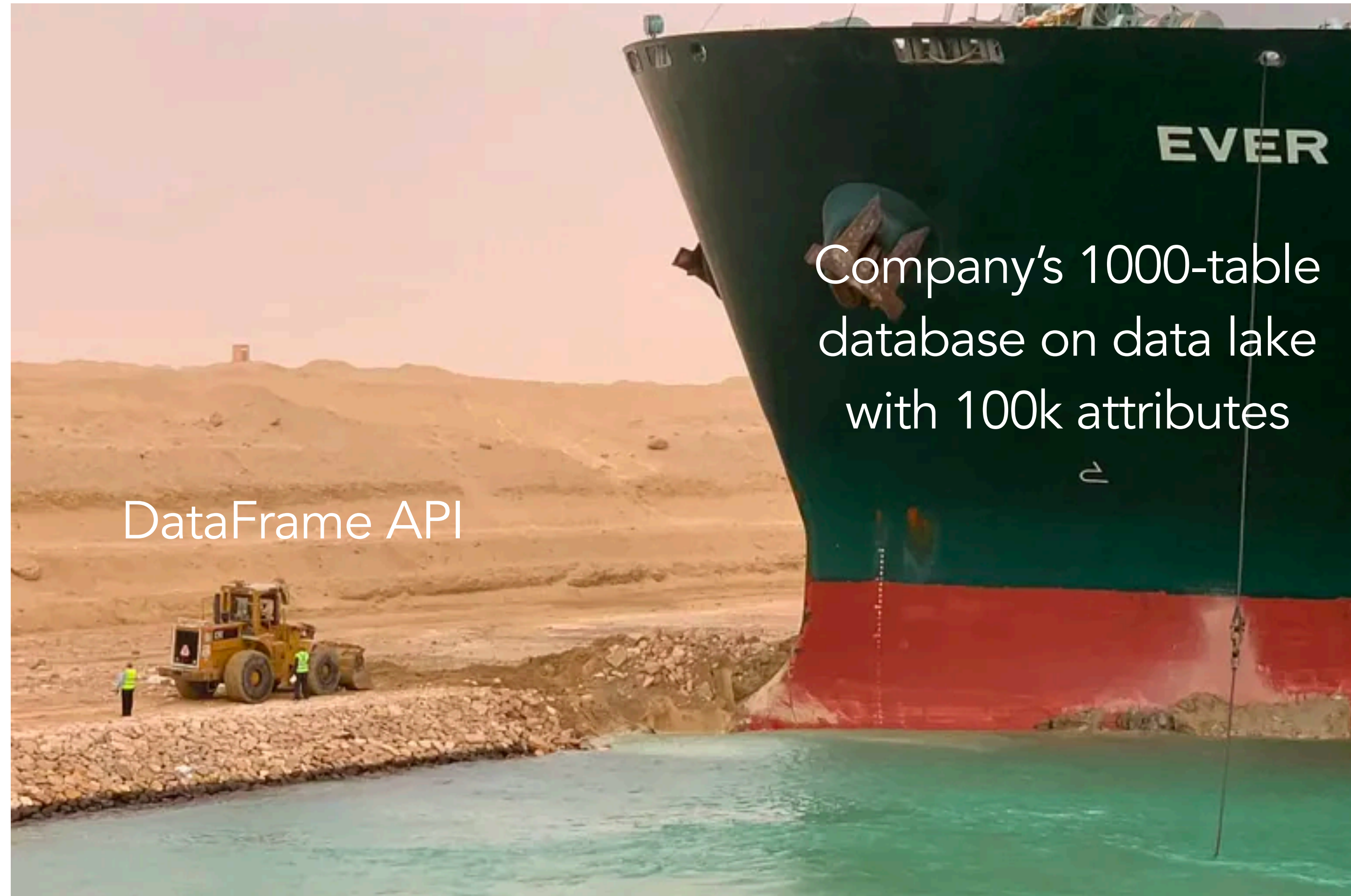


DSC 204a Scalable Data Systems

- Haojian Jin



Company's 1000-table
database on data lake
with 100k attributes

DataFrame API

Today's activities

1. Process.
2. PIA
3. Filesystems and Data Files

Where are we in the class?

Foundations of Data Systems (2.5 weeks)

- Digital representation of Data → Computer Organization → Memory hierarchy → Process → **Storage**

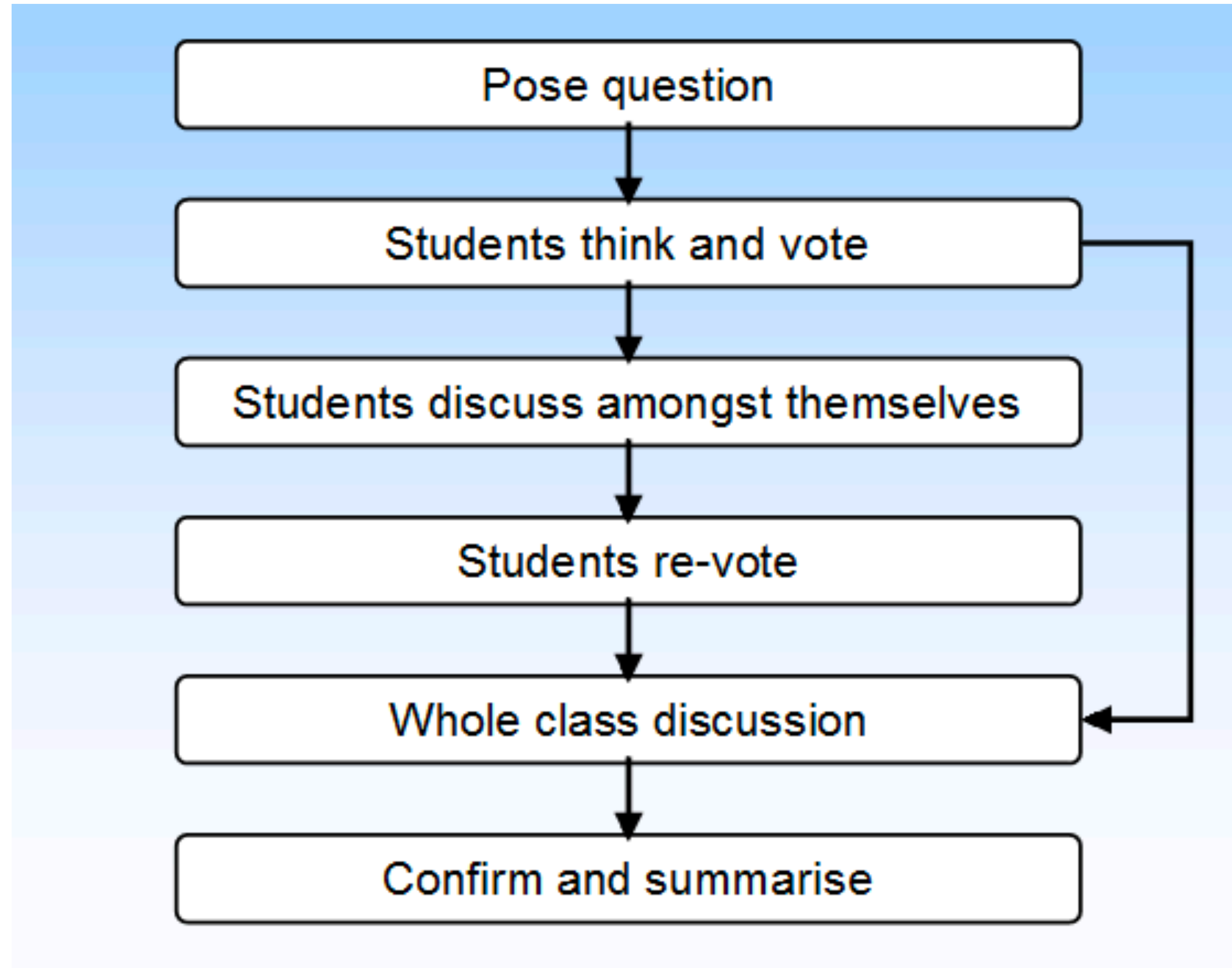
Scaling Distributed Systems (2.5 weeks)

- Cloud → Distributed storage → Partition and replication (HDFS) → Distributed computation

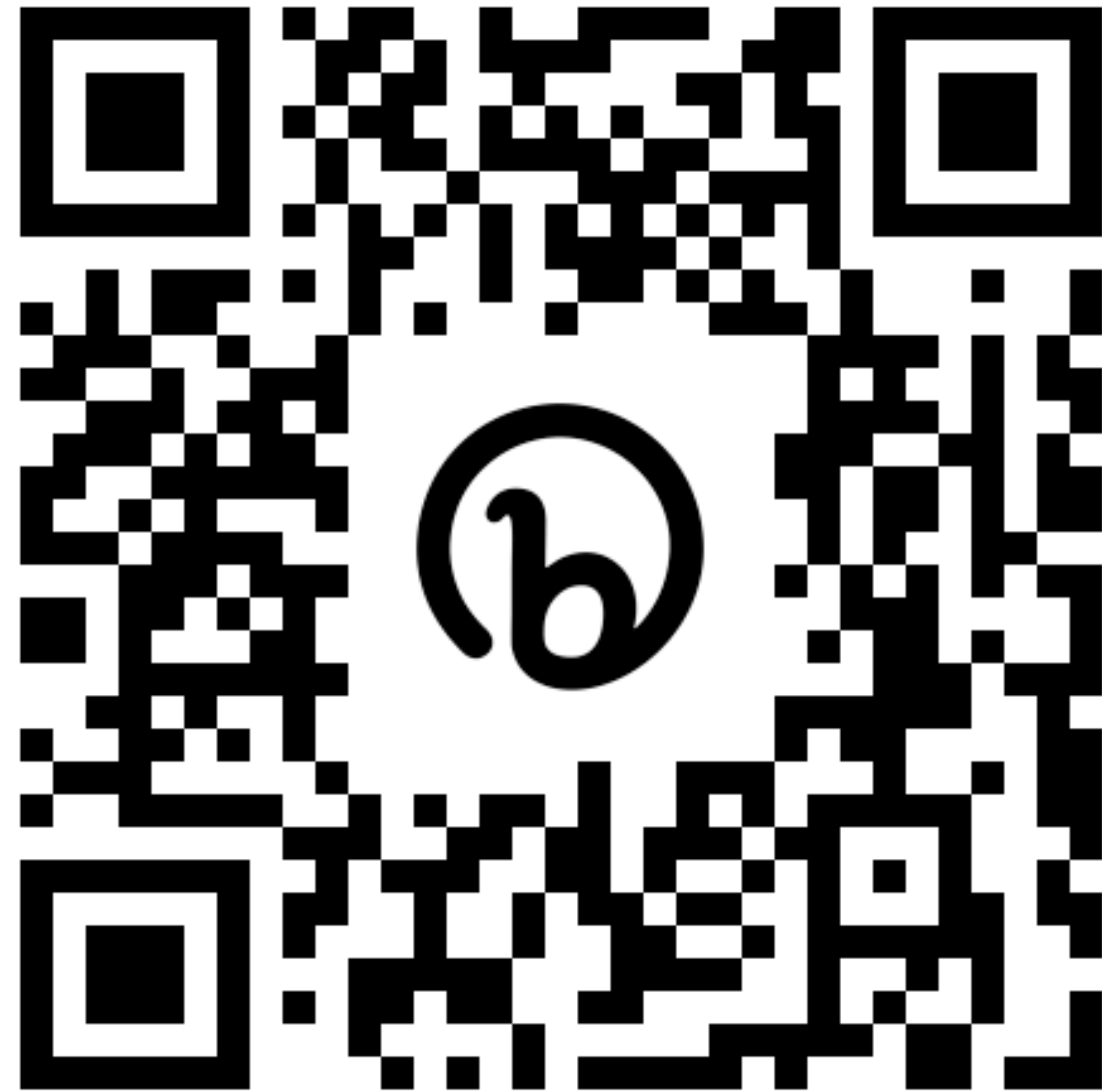
Data Processing and Programming model (5 weeks)

- Data Models evolution → Data encoding evolution → → IO & Unix Pipes → Batch processing (MapReduce) → Stream processing (Spark)

Peer instruction activity



Peer Instruction Activity



bit.ly/dsc204aApr14

Peer Instruction Activity (About 1min per 1pt)

Q1. **[2pts]** Which of these levels of the memory hierarchy typically has the lowest latency to **read** data from?

- A. Flash SSD
- B. Magnetic hard disk
- C. DRAM
- D. All have similar latency
- E. None of the above

C

Peer Instruction Activity (About 1min per 1pt)

Q2. **[2pts]** Which of these levels of the memory hierarchy typically has the lowest latency to **write** data from?

- A. Flash SSD
- B. Magnetic hard disk
- C. DRAM
- D. All have similar latency
- E. None of the above

C

Peer Instruction Activity (About 1min per 1pt)

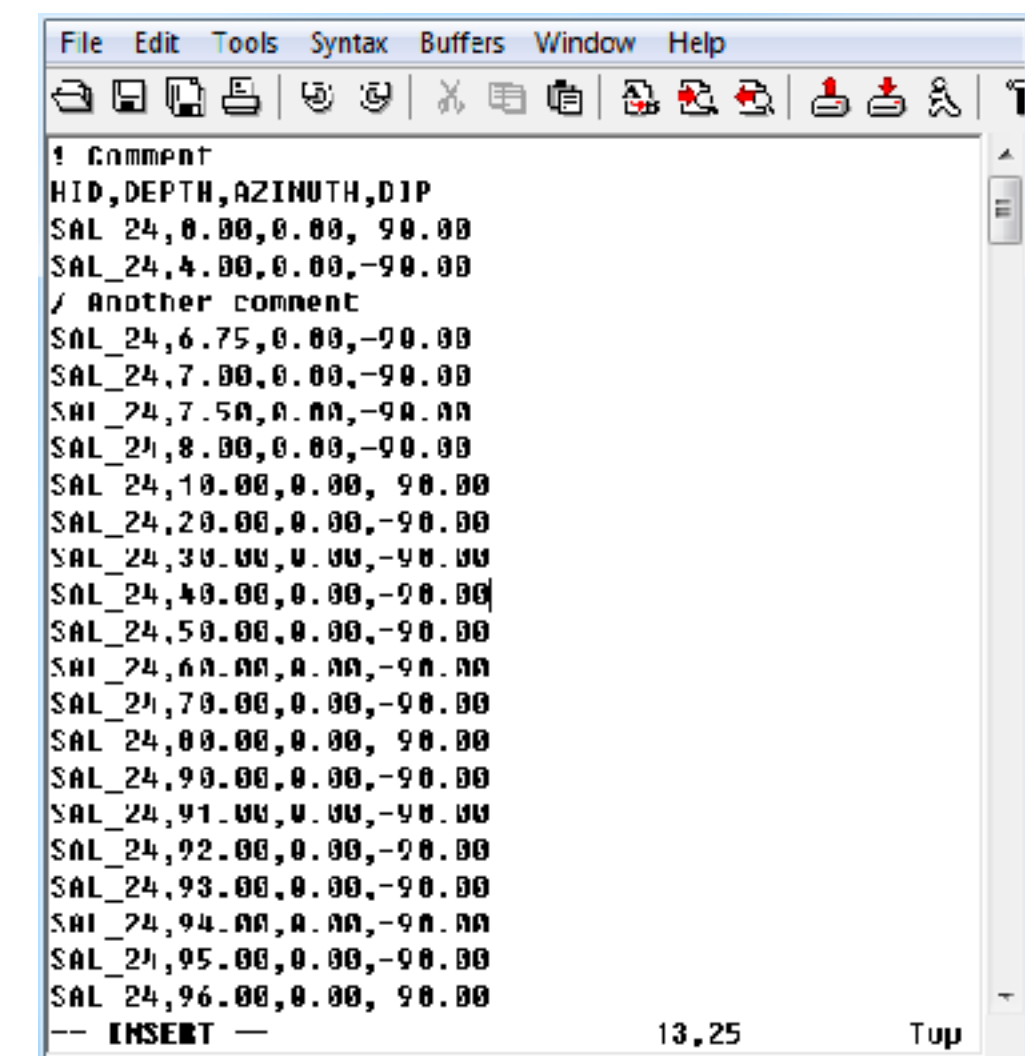
Q3. [6pts] Suppose you have a 750 x 250 matrix of 32-bit integers in Python memory. Its entries are known to be non-negative and < 10 . You write it to disk as a CSV file with one row per line.

What is the rough maximum ratio of the size of your file on disk versus the size of the matrix as an object in DRAM?

The answer is one of the following. If you pick the right answer, you need not explain. If you pick the wrong answer and seek partial credits, please explain your answer succinctly and clearly.

- | | | |
|---------|---------|---------|
| A. 0.25 | B. 0.5 | C. 0.75 |
| D. 1.0 | E. 1.33 | F. 1.5 |
| G. 1.67 | H. 2.0 | |

Hints:



```
File Edit Tools Syntax Buffers Window Help
! Comment
HID,DEPTH,AZIMUTH,DIP
SAL_24,0.00,0.00,90.00
SAL_24,4.00,0.00,-90.00
/ Another comment
SAL_24,6.75,0.00,-90.00
SAL_24,7.00,0.00,-90.00
SAL_24,7.50,0.00,-90.00
SAL_24,8.00,0.00,-90.00
SAL_24,10.00,0.00,90.00
SAL_24,20.00,0.00,-90.00
SAL_24,30.00,0.00,-90.00
SAL_24,40.00,0.00,-90.00
SAL_24,50.00,0.00,-90.00
SAL_24,60.00,0.00,-90.00
SAL_24,70.00,0.00,-90.00
SAL_24,80.00,0.00,90.00
SAL_24,90.00,0.00,-90.00
SAL_24,91.00,0.00,-90.00
SAL_24,92.00,0.00,-90.00
SAL_24,93.00,0.00,-90.00
SAL_24,94.00,0.00,-90.00
SAL_24,95.00,0.00,-90.00
SAL_24,96.00,0.00,90.00
-- INSERT --
```


Peer Instruction Activity (About 1min per 1pt)

Q3. [6pts] Suppose you have a 750 x 250 matrix of 32-bit integers in Python memory. Its entries are known to be non-negative and < 10 . You write it to disk as a CSV file with one row per line.

What is the rough maximum ratio of the size of your file on disk versus the size of the matrix as an object in DRAM?

The answer is one of the following. If you pick the right answer, you need not explain. If you pick the wrong answer and seek partial credits, please explain your answer succinctly and clearly.

- | | | |
|---------|---------|---------|
| A. 0.25 | B. 0.5 | C. 0.75 |
| D. 1.0 | E. 1.33 | F. 1.5 |
| G. 1.67 | H. 2.0 | |

Final answer: B) 0.5

Total count of numbers = $750 * 250 = x$, say (no need to calculate this!)

Size as matrix object in DRAM: $x * 4B$

A single digit integer is 1B as char

CSV means each number has “,” after it, except for the last “\n”; so, 1B per number extra in the CSV file

So, size as file on disk is roughly $x * 2B$

Thus, the max ratio is $x*2 / (x*4) = 0.5$

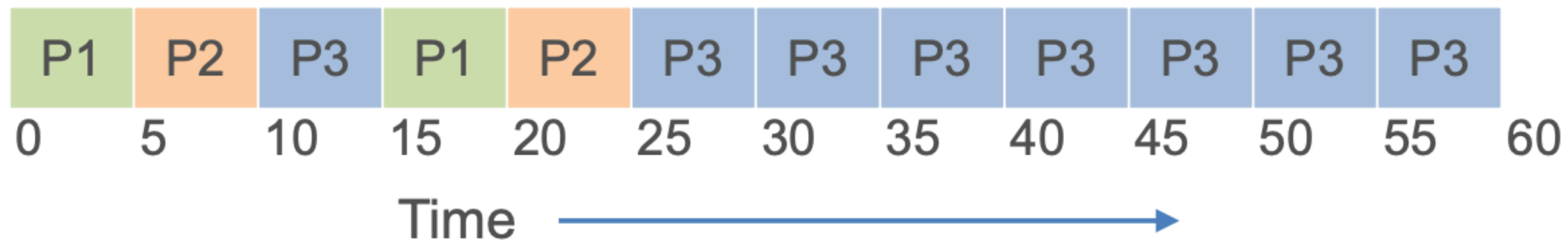
Peer Instruction Activity (About 1min per 1pt)

Q4. **[3x6pts]** Here is a Gantt Chart for 3 processes of the given lengths that all arrive at time 0.

- A) What is the rough *average response time*?
- B) What is the rough *average turnaround time*?

A. 5
B. 35

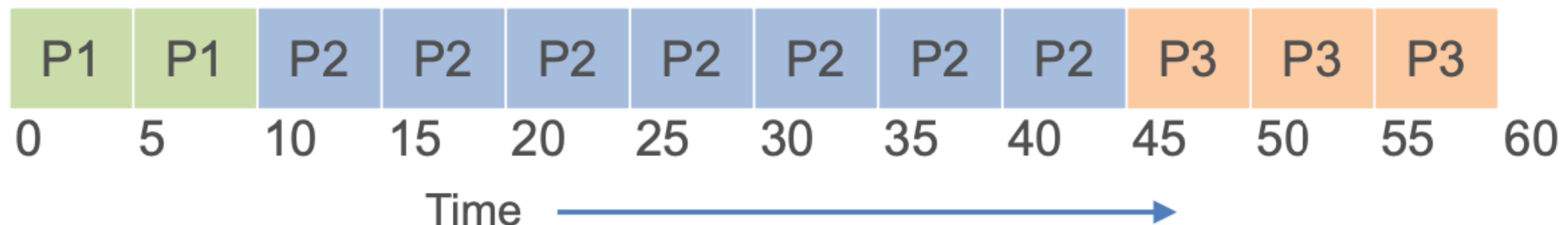
P1, P2, and P3 are of lengths 10, 10, and 40 units, resp.



Peer Instruction Activity (About 1min per 1pt)

Q5. (After class) [3x6pts] Here is a Gantt Chart for 3 processes of the given lengths that arrive at the different times given.

- A) What is the rough *average response time*?
- B) What is the rough *average turnaround time*?
- C) Which scheduling policy/policies discussed in class (FIFO, SJF, P1, P2, and P3 are of lengths 10, 35, and 15 units, resp. and arrive at times 0, 10, and 20, resp.



Peer Instruction Activity (About 1min per 1pt)

Q6. **[3x2pts]** For each of the following answer True or False:

- A) An OS typically has mechanisms to wrest control of hardware back from a user process.
- B) SCTF is the fairest scheduling policy we discussed in class.
- C) CPU caches are usually cheaper per MB than Flash SSD.

A. True
B. False
C. False

Peer Instruction Activity (About 1min per 1pt)

Q7.(**After class**) Suppose you are given that n processes of the same lengths (k units each) all arrive at roughly the same time. Answer these questions.

- A. **[8pts]** Which scheduling policy/policies discussed in class (FIFO, SJF, SCTF, RR) will give the *lowest average turnaround time*? What is that lowest value?
- B. **[8pts]** Which scheduling policy/policies discussed in class (FIFO, SJF, SCTF, RR) will give the *lowest average response time*? What is that lowest value?
- C. **[4pts]** For what k will all 4 policies yield the same metrics?

Today

Process management: virtualization & Concurrency
Filesystems and Data Files

Q: What is a file?



Abstractions: File and Directory

- **File:** A persistent sequence of bytes that stores a logically coherent digital object for an application
 - **File Format:** An application-specific standard that dictates how to interpret and process a file's bytes
 - 100s of file formats exist (e.g., TXT, DOC, GIF, MPEG); varying data models/types, domain-specific, etc.
 - **Metadata:** Summary or organizing info. about file content (aka *payload*) stored with file itself; format-dependent
- **Directory:** A cataloging structure with a list of references to files and/or (recursively) other directories
 - Typically treated as a special kind of file
 - Sub dir., Parent dir., Root dir.

Filesystem

- **Filesystem:** The part of OS that helps programs create, manage, and delete files on disk (sec. storage)
- Roughly split into *logical level* and *physical level*
 - Logical level exposes file and dir. abstractions and offers System Call APIs for file handling
 - Physical level works with disk firmware and moves bytes to/from disk to DRAM

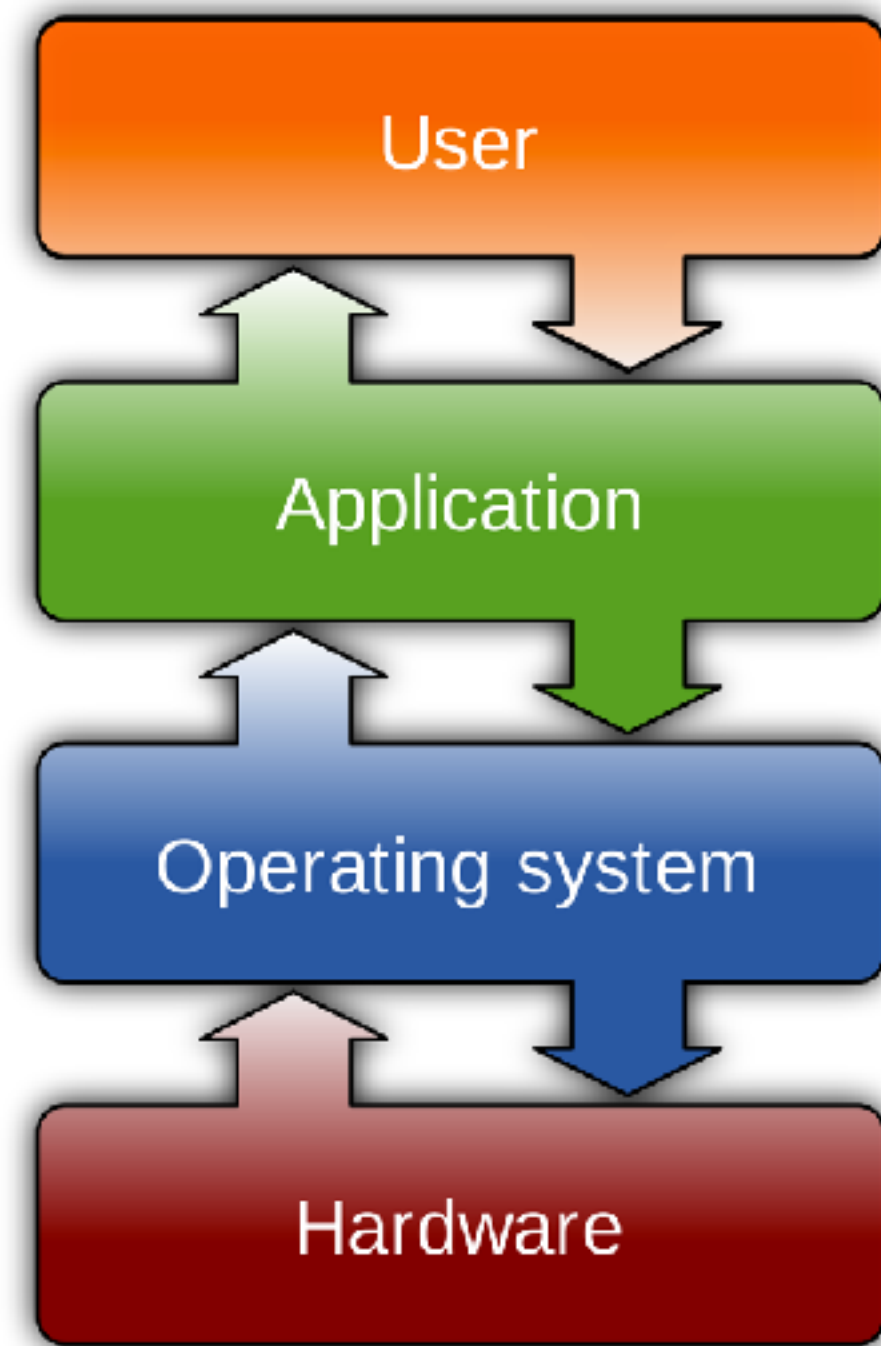
Filesystem

- Dozens of filesystems exist, e.g., ext2, ext3, NTFS, etc.
 - Differ on how they layer file and dir. abstractions as bytes, what metadata is stored, etc.
 - Differ on how data integrity/reliability is assured, support for editing/resizing, compression/encryption, etc.
 - Some can work with (“**mounted**” by) multiple OSs

Virtualization of File on Disk

- OS abstracts a file on disk as a virtual object for processes
- **File Descriptor:** An OS-assigned +ve integer identifier/reference for a file's virtual object that a process can use
 - 0/1/2 reserved for STDIN/STDOUT/STDERR
 - **File Handle:** A PL's abstraction on top of a file descr. (fd)

System Call API for File Handling:



API of OS called "System Calls"

- **open()**: Create a file; assign fd; optionally overwrite
- **read()**: Copy file's bytes on disk to in-mem. buffer; sized
- **write()**: Copy bytes from in-mem. buffer to file on disk
- **fsync()**: "Flush" (force write) "dirty" data to disk
- **close()**: Free up the fd and other OS state info on it
- **lseek()**: Position offset in file's fd (for random R/W later)
- Dozens more (rename, mkdir, chmod, etc.)

Q: What is a database? How is it different from just a bunch of files?

Collection of files?

Virtualization of Files

Binary Representation on
Disk storage

- Maintenance
- Performance
- Usability
- Security & privacy
- ...

Files Vs Databases: Data Model

- **Database:** *An organized* collection of interrelated data
 - **Data Model:** An abstract model to define organization of data in a formal (mathematically precise) way
 - E.g., Relations, XML, Matrices, DataFrames

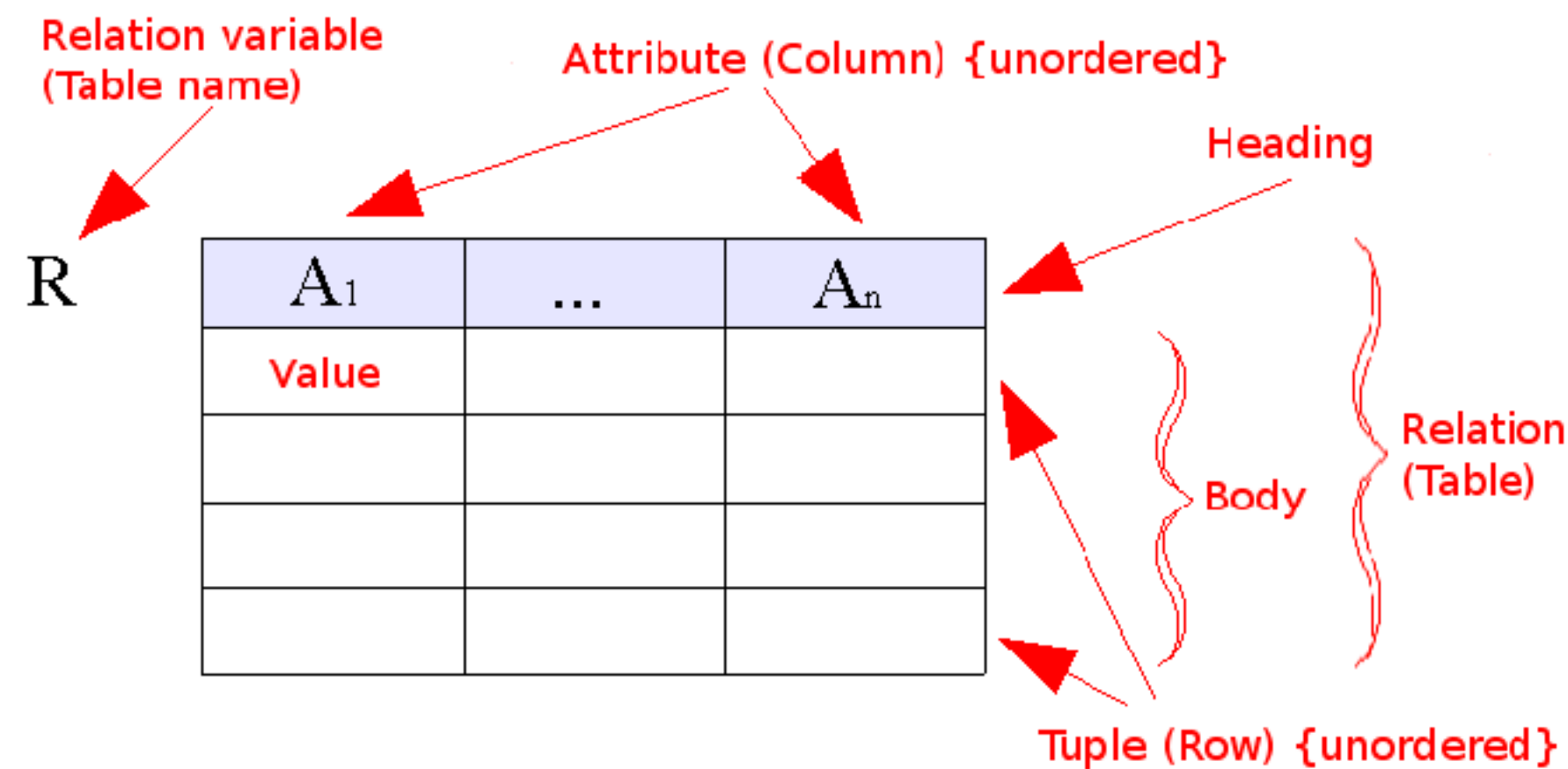
Files Vs Databases: Data Model

- Every database is just an *abstraction* on top of data files!
 - **Logical level:** Data model for higher-level reasoning
 - More in the later lectures.
 - **Physical level:** How bytes are layered on top of files
 - More in the later lectures.
- All data systems (RDBMSs, Dask, Spark, TensorFlow, etc.) are application/platform software that use OS System Call API for handling data files

Data as File: Structured

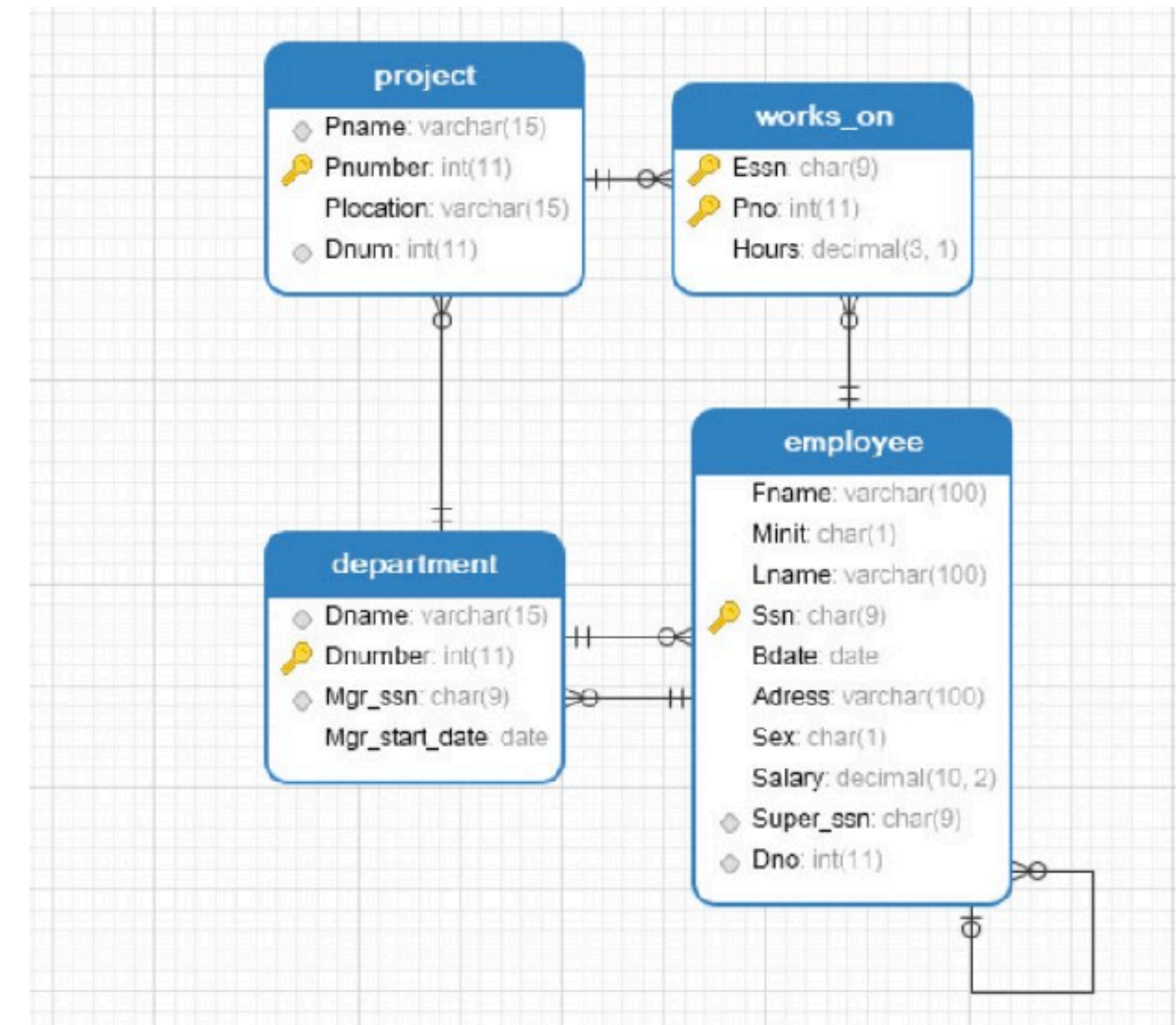
- **Structured Data:** A form of data with regular substructure

Relation



- Most RDBMSs and Spark serialize a relation **as binary file(s)**, often compressed

Relational Database



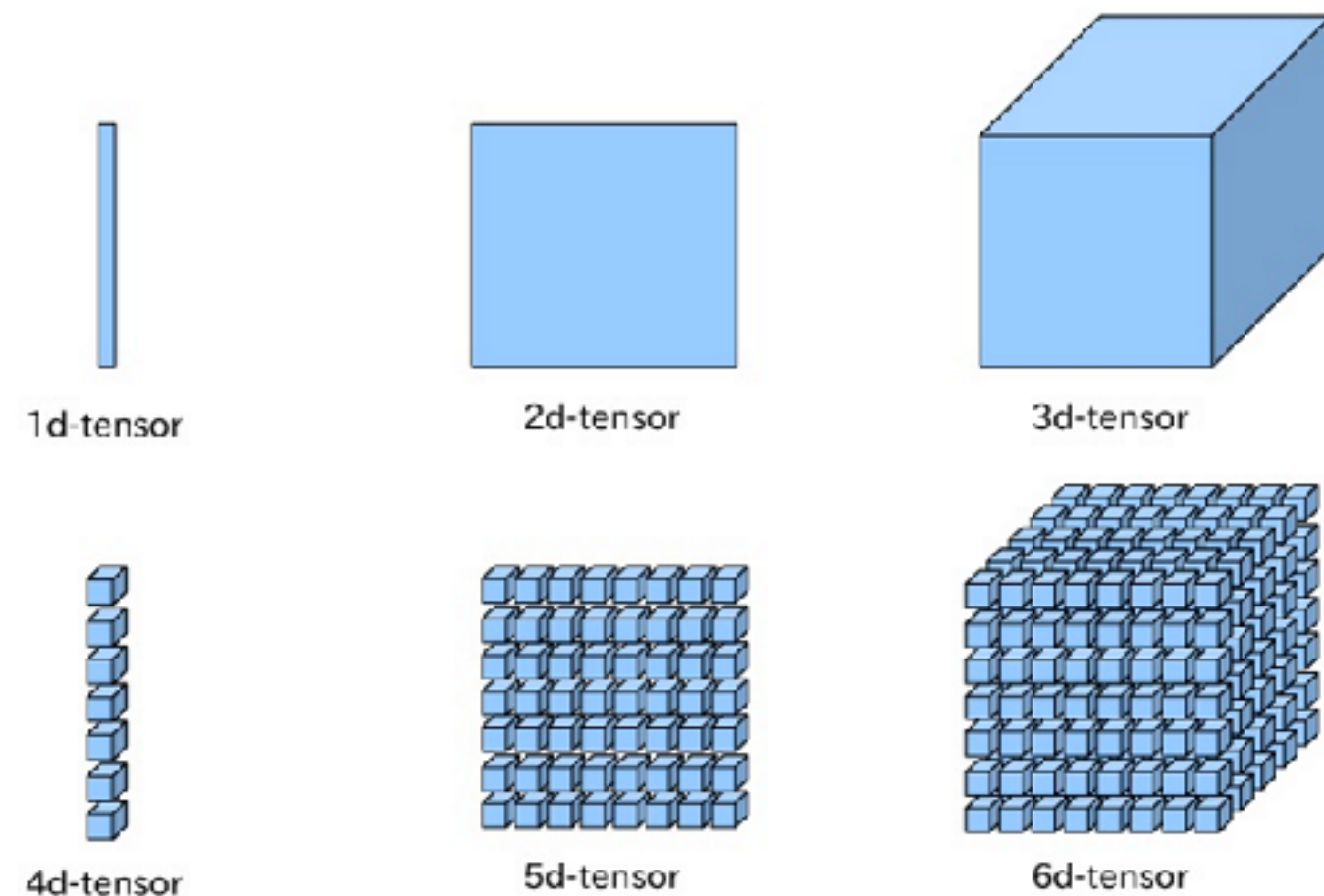
Data as File: Structured

- **Structured Data:** A form of data with regular substructure

Matrix

$$\begin{matrix} & \begin{matrix} 1 & 2 & \dots & n \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ \vdots \\ m \end{matrix} & \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ a_{31} & a_{32} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \end{matrix}$$

Tensor



DataFrame

	Columns			
	Name	Score	Attempts	Qualify
0	Anastasia	12.5	1	yes
1	Dima	9.0	3	no
2	Katherine	16.5	2	yes
3	James	NaN	3	no
4	Emily	9.0	2	no

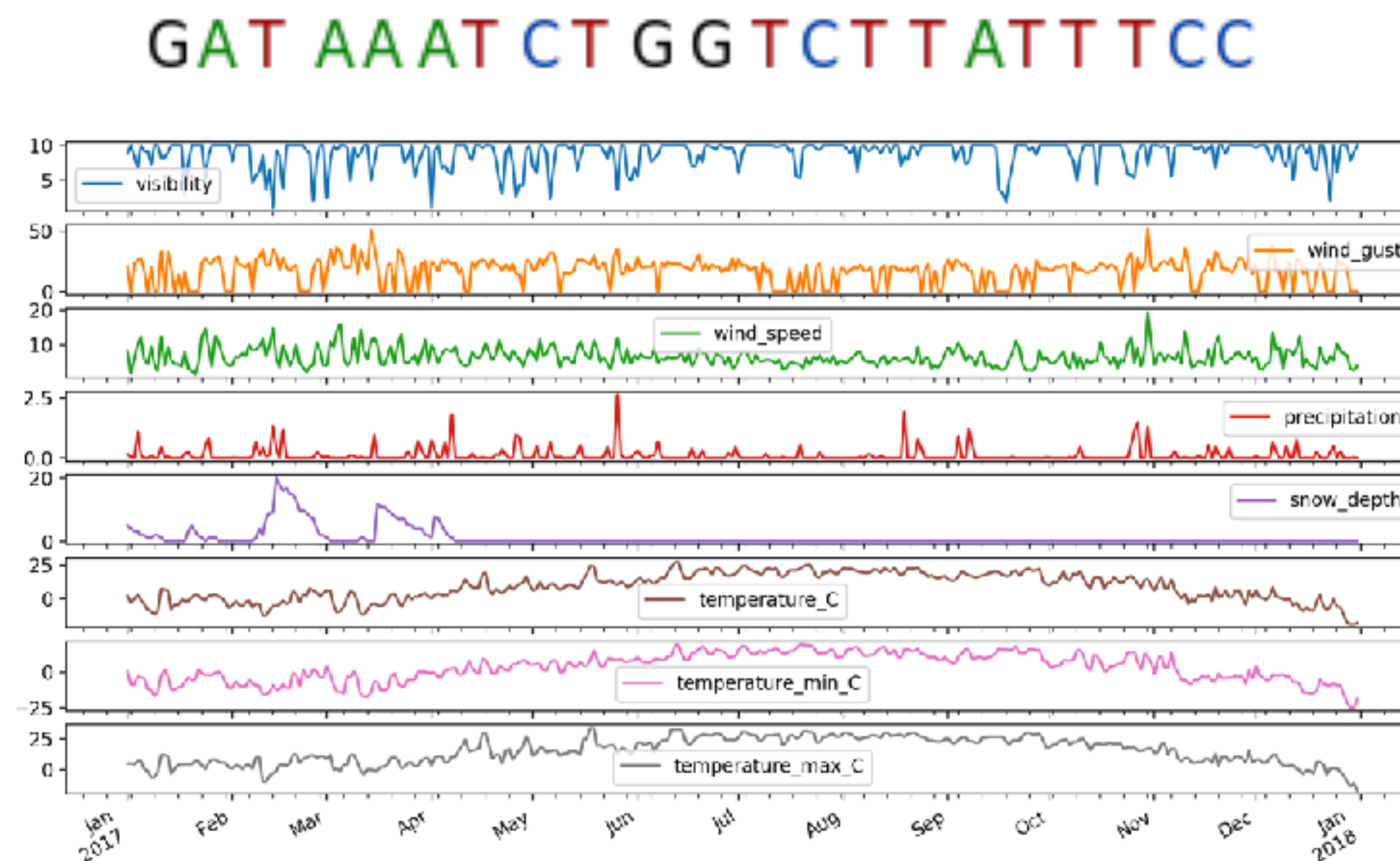
Labels: Rows (on the left), Data (at the bottom right, pointing to the data cells).

- Typically serialized as restricted ASCII text file (TSV, CSV, etc.)
- Matrix/tensor as binary too
- Can layer on Relations too!

Data as File: Structured

- **Structured Data:** A form of data with regular substructure

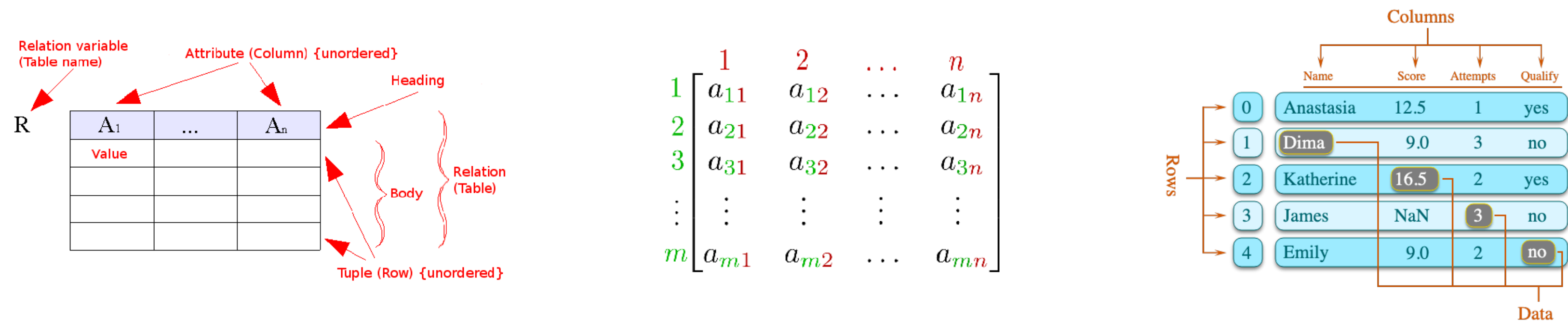
Sequence
(Includes
Time-series)



- Can layer on Relations, Matrices, or DataFrames, or be treated as first-class data model
- Inherits flexibility in file formats (text, binary, etc.)

Comparing Struct. Data Models

Q: What is the difference between Relation, Matrix, and DataFrame?

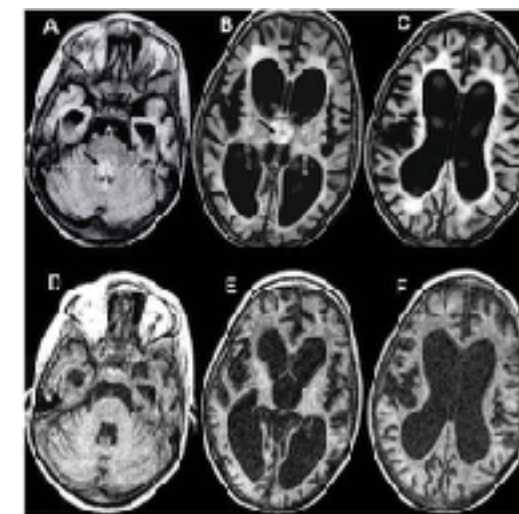
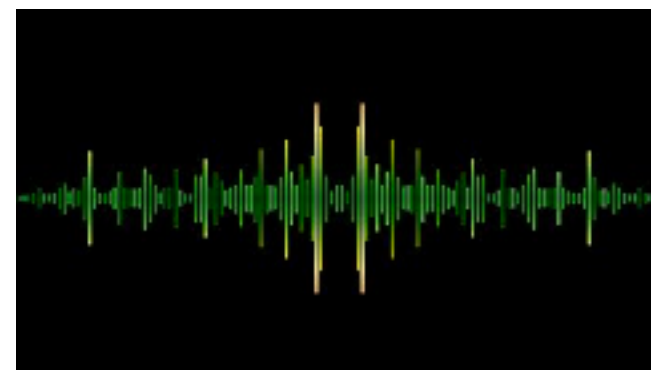


- **Ordering:** Matrix and DataFrame have row/col numbers; Relation is orderless on both axes!
- **Schema Flexibility:** Matrix cells are numbers. Relation tuples conform to pre-defined schema. DataFrame has no pre-defined schema but all rows/cols can have names; col cells can be mixed types!
- **Transpose:** Supported by Matrix & DataFrame, not Relation

If interested in reading more:
<https://towardsdatascience.com/preventing-the-death-of-the-dataframe-8bca1c0f83c8>

Data as File: Other Common Formats

- **Machine Perception** data layer on tensors and/or time-series
- Myriad binary formats, typically with (lossy) compression, e.g., WAV for audio, MP4 for video, etc.



- **Text File** (aka plaintext): Human-readable ASCII characters
- **Docs/Multimodal File**: Myriad app-specific rich binary formats

